# Úloha 6

## Zadání

Vhodnou substitucí převeďte integrál na integrál z racionálních funkcí a ten se pokuste vyřešit:

$$\int \frac{x-\sqrt{x^2+3x+2}}{x+\sqrt{x^2+3x+2}}\,\mathrm{d}x$$

## Řešení

$$\int \frac{x-\sqrt{x^2+3x+2}}{x+\sqrt{x^2+3x+2}}\,\mathrm{d}x$$

$$\sqrt{x^2+3x+2} = -x+t$$

$$x = \frac{t^2-2}{3+2t}$$

$$\mathrm{d}x = \frac{2t^2+6t+4}{(3+2t)^2}\,\mathrm{d}t$$

$$\int \frac{x-\sqrt{x^2+3x+2}}{x+\sqrt{x^2+3x+2}}\,\mathrm{d}x = \int \frac{2\frac{t^2-2}{3+2t}-t}{t}\frac{2t^2+6t+4}{(3+2t)^2}\,\mathrm{d}t =$$

$$\int \frac{(-3t-4)(2t^2+6t+4)}{t(2t+3)^3}\,\mathrm{d}t = -2\int \frac{3t^3+9t^2+6t+4t^2+12t+8}{t(2t+3)^3}\,\mathrm{d}t =$$

$$-2\int \frac{3t^3+13t^2+18t+8}{t(2t+3)^3}\,\mathrm{d}t = -2\int \frac{A}{t}+\frac{B}{2t+3}+\frac{C}{(2t+3)^2}+\frac{D}{(2t+3)^3}\,\mathrm{d}t$$

$$A(2t+3)^3 + Bt(2t+3)^2 + Ct(2t+3) + Dt = 3t^3 + 13t^2 + 18t + 8$$

$$t = 0 \qquad A = \frac{8}{27}$$

$$t = \frac{-3}{2} \qquad D = \frac{-1}{12}$$

$$\frac{64}{27}t^3 + 4Bt^3 = 3t^3$$

$$B = \frac{17}{108}$$

$$\frac{32}{3}t^2 + \frac{17}{9}t^2 + 2Ct^2 = 13t^2$$

$$C = \frac{2}{9}$$

$$-2\int \frac{8}{27t} + \frac{17}{108(2t+3)} + \frac{2}{9(2t+3)^2} - \frac{1}{12(2t+3)^3}\,\mathrm{d}t =$$

$$-\frac{16}{27}\ln|t| - \frac{17}{108}\ln|2t+3| + \frac{2}{9(2t+3)} - \frac{1}{24(2t+3)^2} + C =$$

$$-\frac{16}{27}\ln|t| - \frac{17}{108}\ln|2t+3| + \frac{32t+45}{72(2t+3)^2} + C$$

## Výsledek

$$-\frac{16}{27}\ln|\sqrt{x^2+3x+2}+x| - \frac{17}{108}\ln|2\sqrt{x^2+3x+2}+2x+3| +$$
$$+\frac{32\sqrt{x^2+3x+2}+32x+45}{72(2\sqrt{x^2+3x+2}+2x+3)^2} + C$$